

PAVITTAR SINGH

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Senior Software Engineer · Backend & Full-Stack

Senior software engineer with 6+ years building product-grade backends and full-stack web apps for SaaS and consumer startups. Currently focused on scalable Node/NestJS systems on AWS, with hands-on AI and agentic tooling.

WORK EXPERIENCE

Jul 2025 – Present

Senior Software Engineer

Kuby Ltd. · Cyprus · Remote

- Architected a scalable backend on **NestJS, TypeScript, PostgreSQL, and Prisma**, running on **AWS**.
- Consolidated a standalone MySQL **microservice** (9 tables, 52 files) directly into the core NestJS **modular monolith**, eliminating cross-service auth/API hops and shrinking the deploy surface to one service.
- Shipped and maintain **3 production Next.js portals** (customer, companion, admin) on Vercel; migrated **30+ legacy routes** from a prior member portal via middleware redirects, preserving reseller links and auth continuity.
- Migrated **100+ database-stored emails** into a **182-template Handlebars pipeline** (English/German/Czech) with locale-aware URL substitution, fixing a recurring class of cross-locale broken-link bugs.
- Migrated **213K+ legacy MySQL records** to PostgreSQL/Prisma via custom **ETL** scripts with FK validation and dedup, replacing per-row upserts with bulk loads to cut a multi-hour job to minutes.
- Migrated file storage from EFS to **AWS S3**, cutting **cloud storage costs by 92.5%**.
- Hardened the **CI/CD pipeline** (GitHub Actions → **Docker** → **EC2**), fixing a migration-ordering bug that had caused a **108-minute production outage** — deploys now run schema migrations before container swap for **zero-downtime** releases.
- Built a **CI bundle-budget gate** enforcing a 270 KB per-route JS ceiling on every PR, locking in a **302→113 KB (63%)** code-split win on a high-traffic route and blocking future regressions.
- Authored a reusable, phased **performance-audit framework** (baseline capture, Lighthouse/bundle profiling, guardrail CI checks) as a Claude Code plugin, run across all 3 frontends and the backend.
- Cut Vercel image-optimization costs on the customer portal — **99.9% of the team's total image-transformation billing** — by extending cache TTL to 31 days and trimming the responsive image ladder to actual render widths.
- Stood up a standalone, platform-wide **Playwright E2E suite** plus a manual regression playbook spanning 3 frontends, 2 microservices, and the backend API, serving as the release sign-off gate.
- Introduced **Jest** unit testing across NestJS APIs from zero, growing coverage to **56 spec files** across auth, sessions, webhooks, and companion modules.
- Built internal **AI agents and dev tooling on the Anthropic Claude SDK**, and ran **Claude Code** as a daily IDE pair — accelerating architecture exploration, code review, and refactors.
- Owned **cross-functional communication with founders and product leads** on launch scope, production-crisis triage, and roadmap alignment — and stood up team operating cadence (Slack, Trello, structured release/triage process) for an early-stage engineering function.

Aug 2024 – Jun 2025

Career Break — MCA & AI Upskilling

- Completed **Master of Computer Applications (MCA)** from IGNOU, finishing coursework deferred during prior full-time roles.
- Deep-dived into **agentic frameworks, LLM tooling, and RAG** through hands-on prototypes — directly applied at Kuby.

Aug 2023 – Jul 2024

Senior Software Engineer

Karmasuite · San Francisco, US · Remote

- Drove core development of Karmasuite's **non-profit grant-management SaaS** on **Next.js, tRPC, PostgreSQL (NeonDB), Prisma, and Inngest**.

SKILLS & TOOLING

Backend

Node.js NestJS Express.js tRPC
Prisma PostgreSQL MongoDB MySQL
Inngest Firebase

Frontend

React Next.js Redux Tailwind CSS
Material UI Radix UI React Query
React Hook Form Framer Motion

Languages

TypeScript JavaScript Python SQL
HTML CSS/SCSS

DevOps & Platforms

AWS (S3 EC2 EFS) Docker Vercel
GitHub Actions Git

Architecture & Practices

Microservices Modular Monoliths
System Design Distributed Systems
ETL Pipelines Zero-Downtime Deployments
CI/CD API Design Performance Optimization
E2E Testing

Tools & Testing

Jest Playwright ESLint Prettier Sentry
Jira Linear VSCode

EDUCATION

2022 – 2025

Master of Computer Applications (MCA)

IGNOU · New Delhi, India

2015 – 2018

Bachelor of Computer Applications (BCA)

IGNOU · New Delhi, India

PROJECTS

drill

2026 🔄

AI-powered daily workout app — personalized workouts generated by Claude with exercise swaps and set/rep logging.

Tech: TypeScript, Anthropic Claude SDK

mechadealer

2024 🔄

- Owned the **Fund Management module**, letting organizations create and oversee multiple funds with distinct budgets and objectives.
- Built a fine-grained **Permission Management System** on Clerk + tRPC, enforcing role-based access across pages and APIs.
- Optimized API performance by trimming response payloads, splitting endpoints, and substituting raw SQL where Prisma was a bottleneck.
- Shipped reusable React + Tailwind UI components and AG Grid spreadsheets with save-as-you-edit semantics.

Dec 2021 – Aug 2023

Frontend Engineer (SDE I)

Kenko Health · Bengaluru, India

- Owned **kenkohealth.in** end-to-end on Next.js, TypeScript, MUI, and Redux for the consumer health product.
- Mentored a team of **3 frontend developers**; enforced quality with ESLint, Husky, and Prettier.
- **Improved site speed by ~60%** via SSR, code-splitting, lazy loading, and Lighthouse-driven audits — lifting SEO and reducing bounce.
- **Cut infrastructure cost by ~40%** by moving the Next.js app to AWS via the Serverless Framework.
- Built a B2B internal portal on MUI, react-admin, React Query, and React Hook Form.
- Designed and shipped **Kenko Quiz V3** — a conditional-path health assessment journey serving **50K+ users**.
- Instrumented UTM-tracked user journeys, enabling marketing to measure and optimize campaign funnels.
- Integrated Sentry for real-time error tracking, shortening bug-resolution cycles.

May 2021 – Nov 2021

MERN Stack Developer

Aplore Technologies · India

- Delivered **4 end-to-end web applications** (e-commerce, sales, recruitment) — full-stack ownership from architecture to production.
- Built **JK Sales Smart** on React with **real-time SAP API sync** via Node.js Event API and Socket.io.
- Standardized deployments across projects on **AWS EC2** with NGINX, PM2, and a GitHub deploy pipeline.
- Mentored 3–5 junior developers via code reviews and served as primary technical liaison to clients.
- Used **AWS S3** for secure file storage and Node.js + cron for task automation.

Jul 2020 – Dec 2020

JavaScript Developer

The Loot · Tamil Nadu, India · Freelance

- Launched and maintained *theloot.app*, a full-stack K-12 educational platform.
- Engineered a **user-engagement and Rewards system** on **Firebase Cloud Functions** in a microservice layout.
- Built an internal **React** dashboard for data management, monitoring, and admin decision-making.
- Managed the Firebase backend for secure, efficient data storage and retrieval.

Nov 2019 – Apr 2020

Web Developer

Augment Systems · New Delhi, India · Internship

- Maintained the **Orders Management** project on TypeScript, Angular, and Node.js.
- Built a proprietary **Citation Engine** auto-generating correctly formatted academic citations.
- Assisted with **AWS EC2** deployment and maintenance.

FinTech platform letting retail investors deploy and monitor automated trading strategies with real-time performance metrics.

Tech: Python, FastAPI

editorjs-html

2020 · 379 stars

Open-source library converting Editor.js JSON into clean semantic HTML. 379 GitHub stars, actively maintained, published on npm.

Tech: TypeScript, Rollup, Jest, npm

Conway's Game of Life

2025

Web-based simulation visualizing how complex evolving patterns emerge from simple rules.

Tech: React, TypeScript

BLOGS

Unique IDs... Which is the best?

How consistent hashing scales distributed systems

Deciphering CAP: The fundamental trade-off in distributed systems